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Procedia Computer Science 00 (2018) 000–000

Procedia Computer Science

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The 17th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN 2026)
October 28-30, 2026, Almaty, Kazakhstan

Continuous Behavioral Authentication for Mobile Devices Using an Adaptive LSTM-Based Intelligent Agent

Madi Gali[a,*], Vassiliy Serbin[b], Aray Kassenkhan[c]

[a]*Department of Cybersecurity, Institute of Automation and Information Technologies, Satbayev University, Almaty 050013, Kazakhstan*
[b]*Department of Information Systems, Institute of Automation and Information Technologies, Satbayev University, Almaty 050013, Kazakhstan*
[c]*Department of Software Engineering, Institute of Automation and Information Technologies, Satbayev University, Almaty 050013, Kazakhstan*

Abstract

Traditional mobile authentication mechanisms, such as Personal Identification Numbers (PINs) and passwords, operate as one-time verification methods that fail to protect device sessions after the initial login event. Continuous authentication (CA) addresses this limitation by persistently verifying user identity throughout an active session based on behavioral signals captured by embedded smartphone sensors. In this paper, we propose a novel continuous behavioral authentication system that integrates a two-layer Long Short-Term Memory (LSTM) network with an Adaptive Intelligent Agent for real-time identity verification on mobile devices. The system operates on accelerometer and gyroscope signals collected from the smartphone at 100 Hz, extracting a compact 36-dimensional statistical feature vector from non-overlapping sliding windows of 100 samples. Sequences of 10 consecutive feature vectors are fed into the LSTM network, which produces per-user confidence scores that are processed by the Adaptive Intelligent Agent. The agent maintains a dynamically evolving user behavioral profile using an Exponential Moving Average (EMA) mechanism and issues authentication decisions based on a sliding window of recent scores, enabling the system to accommodate natural behavioral drift without requiring model retraining. The proposed system is evaluated on the publicly available HMOG dataset comprising 100 subjects under a rigorous 5-fold cross-validation protocol with one-vs-all evaluation. Experimental results demonstrate that the proposed LSTM-based adaptive agent achieves an Equal Error Rate (EER) of $1.28\% \pm 0.35\%$ and a classification accuracy of $98.15\% \pm 0.33\%$, outperforming the Support Vector Machine baseline (EER: 6.74%) by a relative margin of 81% and surpassing the performance reported in the original HMOG study and several recent related works on inertial sensor-based continuous authentication. The low False Rejection Rate of 0.70% further demonstrates the practical usability of the proposed system for real-world continuous authentication scenarios.

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Keywords: continuous authentication; behavioral biometrics; long short-term memory; adaptive agent; inertial sensors; accelerometer; gyroscope; mobile security; exponential moving average; sliding window

* Corresponding author.
E-mail address: Gali.M@stud.satbayev.university

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1. Introduction

Smartphones have witnessed a rapid increase in storage and computational resources, transforming them into invaluable instruments for daily activities and the management of sensitive personal information [1]. These devices are ubiquitous in modern life, with global shipments reaching hundreds of millions of units annually [2]. Despite their popularity, smartphones remain highly susceptible to security breaches, particularly if the device is lost, stolen, or seized by an unauthorized party [1]. Traditional access control methods, such as Personal Identification Numbers (PINs), alphanumeric passwords, and fingerprints, serve as the primary line of defense [3]. However, these one-shot authentication methods suffer from significant limitations; they are vulnerable to shoulder surfing, smudge attacks, and smudge-based pattern inference [4]. Furthermore, once an intruder bypasses the initial login event, the device remains fully accessible, as traditional systems fail to provide protection throughout the duration of a usage session [5].

To address the inherent vulnerabilities of one-time verification, continuous authentication (CA)—also known as implicit or passive authentication—has emerged as a dynamic security mechanism [1]. Continuous authentication involves the ongoing validation of a user's identity in real time while the device is in use, operating seamlessly in the background without requiring active user intervention [6]. A promising modality for CA is the utilization of embedded inertial sensors, specifically accelerometers and gyroscopes [7]. These sensors capture unique behavioral biometrics, such as gait signatures and micro-movements, which arise from an individual's neuromuscular coordination and habitual interaction styles [8]. Because these patterns are difficult to replicate and can be monitored unobtrusively, they provide a robust foundation for identifying the legitimate owner throughout a session [9].

Classical machine learning algorithms, including Support Vector Machines (SVM) and Random Forest (RF), have frequently been employed as baselines for modeling these behavioral patterns [6]. While these methods can achieve high accuracy in laboratory settings, they often struggle with behavioral drift—a phenomenon where user biometric profiles naturally evolve over time [10]. Behavioral drift, also defined as concept drift or covariate shift, occurs due to external factors such as fatigue, environmental conditions, or physical state changes [6]. Static models trained on initial data often exhibit performance degradation as the temporal distance from the enrollment phase increases [11]. Consequently, static classifiers fail to provide the long-term reliability required for real-world continuous authentication applications [5].

Long Short-Term Memory (LSTM) networks offer a robust solution to these challenges by effectively capturing the temporal dependencies and dynamic nature of sequential sensor data [12]. LSTM architectures utilize specialized gating mechanisms—including input, output, and forget gates—to manage internal memory states, thereby solving the vanishing gradient problem and enabling the learning of long-range dependencies within time-series signals [13]. Unlike classical methods, LSTMs are particularly adept at modeling the complex dynamics inherent in inertial sensor signals [14]. However, many existing LSTM-based systems remain static in their implementation [5]. These systems typically rely on fixed user templates and lack the necessary adaptive mechanisms to update user profiles in response to the gradual shifts in human behavior [1].

In this paper, we propose a novel continuous behavioral authentication system that integrates a deep LSTM network with an Adaptive Intelligent Agent. To overcome the limitations of static templates and mitigate the impact of behavioral drift, our system utilizes Exponential Moving Average (EMA) to periodically and dynamically update the user behavioral profile as new sensory data is captured. Furthermore, the system employs a sliding window of recent classification scores to make final authentication decisions, introducing a temporal buffer that enhances stability and reduces the likelihood of false rejections caused by momentary behavioral anomalies. The proposed system is evaluated on the publicly available HMOG dataset [9], comprising inertial sensor recordings from 100 subjects performing naturalistic mobile interaction tasks. Through rigorous 5-fold cross-validation, our adaptive system achieved an EER of 1.28% and an accuracy of 98.15%, demonstrating superior performance compared to non-adaptive baselines.

The major contributions of this work are summarized as follows:

- We design an adaptive continuous authentication framework that combines deep sequence modeling via a two-layer LSTM network with an Adaptive Intelligent Agent for real-time behavioral identity verification on mobile devices.

- We implement a dynamic user profile update mechanism based on Exponential Moving Average (EMA) that successfully addresses behavioral drift and concept drift in inertial sensor biometric data without requiring model retraining.
- We propose a sliding window decision mechanism over recent authentication scores that minimizes false rejection rates and improves overall system robustness against momentary behavioral anomalies.
- We conduct an extensive experimental evaluation on the HMOG dataset with 100 subjects using 5-fold cross-validation, achieving an EER of 1.28% and accuracy of 98.15% for motion-based continuous authentication.

The remainder of this paper is organized as follows. Section 2 reviews related work on behavioral biometrics and continuous authentication systems. Section 3 describes the HMOG dataset and preprocessing pipeline. Section 4 details the proposed LSTM model and adaptive agent mechanism. Section 5 presents experimental results. Section 6 discusses findings and limitations. Section 7 concludes the paper.

2. Related Work

2.1. Behavioral Biometrics for Mobile Authentication

Behavioral biometrics identify or verify individuals based on quantifiable patterns in human behavior, such as how a user types, walks, or interacts with a touch screen [2]. These modalities are preferred for continuous authentication because they can be captured unobtrusively in the background without requiring active user cooperation [1]. Several recent surveys have systematized this growing body of work. Abuhamad et al. [15] reviewed more than 140 sensor-based continuous authentication approaches spanning motion, gait, keystroke, and touch-gesture modalities, while Rayani and Changder [16] provided a comprehensive taxonomy of continuous authentication methods together with an overview of publicly available evaluation datasets, including HMOG. From an artificial-intelligence perspective, Liang et al. [17] characterized the sensing, feature-extraction, and learning components common to Internet-of-Things continuous-authentication pipelines and highlighted the scarcity of systems that combine deep sequence models with adaptive profile management, a gap that the present work directly addresses.

Keystroke dynamics identify users by their unique typing rhythm, primarily analyzing the hold time of keys and the inter-key latency between successive presses [18]. Researchers have achieved accuracies ranging from 87% to 98% using keystroke dynamics on mobile devices [18]. More recently, deep sequence models have substantially improved free-text keystroke verification: Acien et al. [19] proposed TypeNet, a Long Short-Term Memory-based Siamese architecture trained on tens of thousands of users that achieves competitive equal error rates at web scale, while Al-Saraireh and AlJa'afreh [20] showed that fusing keystroke and swipe-gesture biometrics further reduces error rates compared with either modality alone.

Touch-based authentication focuses on analyzing the spatiotemporal features of strokes, including pressure, trajectory, and area of contact [11]. The Touchalytics system achieved EERs between 0% and 4% depending on whether the authentication was intra-session or inter-session [11]. Estrela et al. [21] extended this line of work with a continuous-authentication framework for mobile banking that fuses touch-dynamics scores across multiple interaction sessions, and Zhang et al. [22] demonstrated that combining touch interaction patterns with contextual features improves robustness against zero-effort impostor attacks.

Gait recognition utilizes the distinct walking patterns of an individual, often measured via built-in motion sensors while the device is in a user's pocket or hand [23]. While gait-based methods have achieved EERs as low as 0.13%, their performance can be negatively affected by factors such as footwear, fatigue, or uneven ground [5]. Deep-learning architectures have markedly improved gait-based verification in recent years. Gadaleta and Rossi [24] introduced IDNet, which combines convolutional feature extraction with a one-class support vector machine over successive walking cycles; Zou et al. [25] proposed a hybrid CNN-RNN architecture that recognizes gait "in the wild" without constraining the walking route or speed; Giorgi et al. [26] applied recurrent neural networks to raw inertial gait signals with a sliding-window augmentation strategy to increase authentication accuracy; and Choi et al. [27] reduced the gait-capture window to under two seconds while maintaining an accuracy above 90% using a convolutional model deployed directly on the smartphone. Multimodal systems that fuse behavioral traits have been proposed to enhance security and achieve lower error rates than single-modality systems [18, 28], with Cheng et al. [28] recently demonstrating that

an ensemble of CNN, LSTM, and GRU sub-models operating on touch, sliding, and inertial telemetry can be aligned with the Zero Trust Architecture principle of continuous, never-trust-always-verify authentication.

2.2. Inertial Sensor-Based Authentication

Modern smartphones are equipped with inertial sensors, specifically accelerometers and gyroscopes, which provide a rich data source for modeling user-specific interaction patterns [2]. The accelerometer measures linear acceleration forces along three physical axes, while the gyroscope records the angular velocity or rate of rotation [7]. These sensors are particularly valuable for continuous authentication because they do not require specific user permissions in many mobile operating systems and can be sampled at high frequencies [1]. Feature extraction from these signals typically involves constructing descriptive statistical features such as mean, variance, and skewness, alongside frequency-domain features [7]. The SensorAuth system utilized data augmentation techniques to create additional training data from accelerometer and gyroscope signals, achieving a median EER of 4.66% [29]. The HMOG framework captures subtle hand micro-movements during typing, achieving EERs as low as 7.16% [9]. In the DAKOTA system designed for mobile banking, researchers combined touch screen data with motion sensor readings to achieve a 3.5% EER [6]. Studies have shown that gyroscope and orientation features often contribute more significantly to authentication performance than magnetometer data [7]. Convolutional and recurrent architectures applied directly to raw accelerometer and gyroscope streams, such as IDNet [24] and the recurrent gait model of Giorgi et al. [26], have further shown that inertial signals alone, without touch or keystroke data, can support sub-1% EER when combined with sequence-aware deep learning, motivating the sequence-modeling approach adopted in Section 4.

2.3. Classical Machine Learning Approaches

Classical machine learning algorithms, including SVM, RF, and k-Nearest Neighbors (kNN), have been extensively employed as the classification backbone for behavioral biometrics [6]. The Support Vector Machine, originally formulated by Cortes and Vapnik [30], constructs a maximum-margin decision boundary and remains a common linear/kernel baseline in the behavioral-biometrics literature owing to its strong performance on relatively small, high-dimensional feature sets. The Random Forest ensemble, introduced by Breiman [31], aggregates the predictions of many de-correlated decision trees and has repeatedly been reported to outperform single classifiers on tabular, sensor-derived features while remaining computationally inexpensive to train. SilentSense utilized SVM classification to identify users through touch and movement biometrics, achieving 100% accuracy within 12 steps [32]. Random Forest has frequently emerged as a top-performing classifier for gesture-based authentication, with some studies reporting accuracies above 99% [1]. Despite their high accuracy in controlled settings, these classical models often fail to account for behavioral drift [5]. Behavioral drift refers to the natural evolution of a user's biometric profile over time due to external factors such as physical state, environmental conditions, or psychological changes [10]. Static models trained on a fixed enrollment phase often exhibit performance degradation as the temporal distance from the training data increases [11]. Consequently, these classical approaches often require frequent manual re-enrollment or suffer from high false rejection rates in long-term deployments [5]. These well-documented limitations of static classical classifiers directly motivate their use as fixed, non-adaptive baselines in Section 5, against which the proposed adaptive LSTM agent is benchmarked.

2.4. Deep Learning and LSTM for Continuous Authentication

Deep learning architectures, particularly RNNs and LSTM networks, have been proposed to overcome the limitations of classical methods by effectively modeling the temporal dependencies in sensor data [12]. Standard RNNs often suffer from the vanishing gradient problem during long-sequence training [33]. LSTM networks address this by using a specialized cell structure with input, output, and forget gates [13]. The AUToSen framework utilized an LSTM-based model to perform deep sequence learning on smartphone sensor data, achieving an EER of 0.41% [1]. Hybrid CNN-LSTM architectures have also been developed, where a CNN acts as an automatic feature extractor and the LSTM layer serves as a temporal classifier [14]. These hybrid approaches have demonstrated the ability to improve accuracy by nearly 4% compared to traditional SVM-based models [14]. Sequence-modeling approaches have also proven effective outside the inertial-sensor domain: TypeNet [19] applies stacked LSTM layers with a Siamese

loss to free-text keystroke sequences, and Giorgi et al. [26] use recurrent networks over gait cycles extracted from accelerometer and gyroscope signals. Zou et al. [25] combine convolutional and recurrent branches to jointly capture spatial and temporal structure in unconstrained inertial gait data, while Cheng et al. [28] employ an ensemble of CNN, LSTM, and GRU sub-networks across touch, sliding, and sensor modalities to address class imbalance in continuous authentication. All deep models in this study, including the proposed two-layer LSTM, are trained with the Adam optimizer [34], which combines adaptive per-parameter learning rates with momentum and is widely adopted for training recurrent architectures on non-stationary, noisy sensor data.

2.5. Adaptive Authentication Systems

The challenge of maintaining authentication accuracy over time has led to the development of adaptive systems that update user profiles in response to changing behavior [10]. Verma et al. proposed a two-step adaptive user verification algorithm that verifies identity using motion-based biometrics while adapting to changes in human behavior over time [10]. The DAKOTA system incorporated adaptive model updating with new data to address the performance drop caused by the temporal distance between enrollment and authentication [6]. The underlying phenomenon that motivates such adaptive mechanisms is well studied in the broader machine-learning literature under the term *concept drift*, defined as a change in the statistical properties of the target variable over time [35]. Among the techniques surveyed by Gama et al. [35], exponentially weighted moving averages have specifically been proposed for tracking a classifier's operating point as the underlying data distribution evolves: Ross et al. [36] introduced an EWMA-based chart that monitors a streaming classifier's error rate and flags drift once the smoothed estimate departs from its historical baseline, a mechanism conceptually related to the EMA-based profile update employed by the Adaptive Intelligent Agent proposed in Section 4. Recent surveys of continuous authentication [16, 17] similarly identify the lack of lightweight, real-time adaptation mechanisms as an open challenge, since most deep-learning-based systems, including the LSTM and CNN-LSTM baselines reviewed above, still rely on fixed templates learned once during enrollment. However, despite these advancements, most existing LSTM-based systems remain static and lack a robust intelligent agent capable of autonomously updating user profiles in real time [5]. Our work fills this gap by proposing an Adaptive Intelligent Agent that utilizes EMA to autonomously update user profiles and a sliding window of recent authentication scores for robust decision-making.

2.6. Research Methodology

This subsection justifies the principal methodological choices made in the design of the proposed system, namely the selection of the HMOG dataset, the sliding-window configuration, the EMA-based adaptation mechanism, the evaluation protocol, and the classical baselines used for comparison.

Dataset selection. The HMOG dataset was selected over alternative behavioral-biometric corpora for three reasons. First, it is one of the few publicly available datasets that jointly records accelerometer and gyroscope signals from a realistic population of subjects performing naturalistic smartphone tasks, reading, typing, and map navigation, under both sitting and walking conditions, rather than a single constrained activity [9]. Second, HMOG has become a de facto benchmark in the continuous-authentication literature, which allows the results reported here to be directly compared against a substantial body of prior work, including the surveys of Rayani and Changder [16] and Abuhamad et al. [15], both of which list HMOG among the most widely adopted evaluation datasets for inertial-sensor-based authentication. Third, because HMOG subjects perform multiple activities across many sessions, the dataset naturally contains the kind of intra-user behavioral variability that a system must tolerate if it is to remain usable over long deployment periods, precisely the variability that the proposed Adaptive Intelligent Agent is designed to accommodate.

Window size. A non-overlapping window of $W = 100$ samples was chosen because, at the 100 Hz sampling rate used by HMOG, this corresponds to approximately one second of continuous motion, long enough to capture a complete micro-movement pattern (e.g., a hand tremor cycle or a partial gait cycle) while remaining short enough to preserve the fine-grained temporal resolution needed for real-time authentication decisions. Prior inertial-sensor studies report that statistical features computed over comparable one-second windows retain strong discriminative power for user identity [7], and a similar order of magnitude has been used in gait-cycle-based systems such as IDNet [24]. Grouping $L = 10$ consecutive windows into a sequence further allows the LSTM to observe roughly ten seconds

of behavioral context per authentication decision, which is consistent with the sliding-window durations used by recurrent gait-authentication systems [26, 25].

Exponential Moving Average for adaptation. EMA was selected as the profile-update mechanism instead of full model retraining or a fixed enrollment template because it satisfies three practical constraints of on-device continuous authentication: it requires only constant memory (a single running scalar per user), it can be updated with a single multiply-add operation per new score, and it inherently discounts older observations without requiring an explicit forgetting schedule to be learned. This behavior is grounded in the concept-drift literature: Ross et al. [36] demonstrated that an EWMA-based monitor can track a classifier's error rate under gradual distributional change with substantially less variance than a simple sliding-window average, and Gama et al. [35] identify EMA-based schemes as one of the principal families of adaptive-learning techniques for non-stationary data streams. The learning rate $\alpha = 0.05$ was chosen empirically to bias the profile toward long-term behavioral stability, since a larger α was observed to make the profile overly sensitive to momentary anomalies (e.g., typing while standing on a moving bus), while a smaller α slowed adaptation to genuine, sustained behavioral drift.

Cross-validation protocol. A 5-fold cross-validation protocol was adopted, with a one-vs-all evaluation performed independently for every subject in each fold, because it provides a stable estimate of generalization performance while keeping the computational cost of retraining per-subject classifiers tractable. Five folds represent a widely used compromise in the behavioral-biometrics literature between the bias of a small number of folds (e.g., a single train/test split) and the variance and computational overhead of a larger number of folds or leave-one-subject-out evaluation; comparable protocols are used throughout the studies summarized in Table 5, which facilitates a fair comparison of the reported EER values.

Baseline classifier selection. SVM and RF were selected as classical baselines because they represent two structurally different and well-established approaches to the same classification problem, and because both have a long track record as behavioral-biometrics classifiers in the literature reviewed in Section 2C. The Support Vector Machine [30] constructs a single global decision boundary and is representative of the linear/kernel-based classifiers used in early behavioral-biometrics work such as SilentSense [32]; it therefore serves as a lower-complexity, interpretable reference point. The Random Forest [31] is an ensemble method that has consistently achieved among the highest reported accuracies for gesture- and sensor-based authentication [1], making it a stronger, though still non-adaptive, point of comparison. Evaluating the proposed LSTM-based adaptive agent against both a weak and a strong static baseline allows the contribution of sequence modeling and adaptive profiling to be assessed separately, as discussed in Section 6.

3. Dataset and Preprocessing

3.1. HMOG Dataset

The experiments are conducted using the Hand Movement, Orientation, and Grasp (HMOG) dataset, a publicly available benchmark specifically designed for continuous behavioral authentication on mobile devices [9]. The dataset was collected from 100 participants who performed naturalistic tasks on a Samsung Galaxy S3 smartphone, including reading, typing, and map navigation, both while sitting and while walking. Each participant completed 24 sessions, resulting in approximately 2–6 hours of continuous sensor recordings per subject. The HMOG dataset has been widely adopted in the behavioral biometrics research community as a standard benchmark for evaluating continuous authentication systems [9].

3.2. Sensor Modalities

We utilize two inertial sensor modalities: the accelerometer and the gyroscope. The accelerometer measures the linear acceleration of the device along three orthogonal axes (x, y, z), capturing hand movement and micro-motion patterns. The gyroscope measures the angular velocity along the same three axes, capturing rotational movements and orientation changes. These two sensors together provide six data channels, offering a complementary representation of both translational and rotational behavioral dynamics. Magnetometer data was excluded as prior research has

demonstrated that accelerometer and gyroscope features provide more stable and discriminative behavioral signatures for authentication [7].

3.3. Sliding Window Segmentation

Raw sensor streams are segmented using a sliding window of $W = 100$ consecutive sensor readings (approximately one second at 100 Hz). A non-overlapping step size of $S = 100$ samples ensures independence between consecutive windows. A maximum of $N_{max} = 150$ windows are extracted per subject, yielding 15,000 windows across 100 subjects.

3.4. Feature Extraction

From each window, a 36-dimensional feature vector is extracted by computing six statistical descriptors for each of the six sensor channels: mean (μ), standard deviation (σ), minimum, maximum, root mean square (RMS), and skewness. The RMS is defined as:

$$\text{RMS} = \sqrt{\frac{1}{W}\sum_{i=1}^{W} x_i^2} \tag{1}$$

3.5. Data Normalization

All feature vectors are standardized using z-score normalization:

$$\hat{x}_j = \frac{x_j - \mu_j}{\sigma_j}, \quad j = 1, 2, \ldots, 36 \tag{2}$$

where μ_j and σ_j are computed over the training set.

3.6. Sequence Construction for LSTM

Feature vectors are grouped into sequences of $L = 10$ consecutive windows, yielding input tensors of shape $(10, 36)$ per sample. A sliding step of $S_{seq} = 5$ windows is applied between consecutive sequences during training.

3.7. Evaluation Protocol

A one-vs-all authentication protocol is adopted: for each subject, that subject is the legitimate user and all others are impostors. Performance is averaged across all subjects and evaluated under 5-fold cross-validation. Reported metrics are EER, FAR, FRR, and Accuracy as mean ± std across folds.

4. Proposed Method

4.1. System Overview

The proposed system consists of five functional layers forming an end-to-end pipeline from raw sensor acquisition to adaptive authentication decision-making. The pipeline operates continuously in the background during an active device session, requiring no explicit user interaction.

4.2. LSTM Classification Layer

The sequence tensor $(10, 36)$ is fed into a two-layer LSTM network:

$$\mathbf{h}_t = \text{LSTM}_1(\mathbf{x}_t, \mathbf{h}_{t-1}, \mathbf{c}_{t-1}) \tag{3}$$

$$\mathbf{o} = \text{LSTM}_2(\mathbf{h}_t) \tag{4}$$

The first LSTM layer contains 64 hidden units and returns the full output sequence, followed by Dropout ($p = 0.3$). The second LSTM layer contains 32 hidden units and returns only the final hidden state, followed by Batch Normalization. A Dense layer with 32 units and ReLU activation maps the output to an intermediate representation, followed by a second Dropout layer. The Softmax output layer produces a probability distribution over all N user classes:

$$\hat{y}_k = \frac{e^{z_k}}{\sum_{j=1}^{N} e^{z_j}}, \quad k = 1, 2, \ldots, N \tag{5}$$

4.3. Adaptive Intelligent Agent

4.3.1. Profile Updating via EMA

The agent maintains a user profile score p_t updated at each step:

$$p_t = (1 - \alpha) \cdot p_{t-1} + \alpha \cdot s_t \tag{6}$$

where $\alpha = 0.05$ is the profile learning rate and $s_t \in [0, 1]$ is the confidence score from the LSTM.

4.3.2. Decision-Making via Sliding Window

The agent maintains a sliding window buffer of the $W = 10$ most recent scores. The authentication decision is:

$$d_t = \begin{cases} \text{Authenticated} & \text{if } \bar{s}_t \geq \tau \\ \text{Rejected} & \text{if } \bar{s}_t < \tau \end{cases} \tag{7}$$

where $\bar{s}_t$ is the mean of the sliding window and $\tau = 0.5$ is the decision threshold [29].

4.4. LSTM Model Training

The model is trained using categorical cross-entropy loss:

$$\mathcal{L} = -\sum_{k=1}^{N} y_k \log(\hat{y}_k) \tag{8}$$

The Adam optimizer is used with initial learning rate $\eta = 0.001$, reduced by factor 0.5 on plateau. Early Stopping with patience of 5 epochs prevents overfitting. Batch size is 32 and maximum epochs is 50. Table 1 summarizes all hyperparameters.

Table 1. Hyperparameters of the Proposed System

Hyperparameter	Value
Window size (W)	100 samples
Feature vector size (d)	36
Sequence length (L)	10 windows
LSTM layer 1 units	64
LSTM layer 2 units	32
Dense layer units	32
Dropout rate (p)	0.3
Optimizer	Adam
Initial learning rate (η)	0.001
Batch size	32
Max epochs	50
Early stopping patience	5 epochs
Agent learning rate (α)	0.05
Agent window size	10 scores
Decision threshold (τ)	0.5

5. Experiments and Results

5.1. Experimental Setup

All experiments are conducted on the HMOG dataset using 100 subjects with a maximum of 150 non-overlapping windows per subject, yielding 15,000 feature vectors. The proposed LSTM agent is compared against two baselines: LinearSVC calibrated with 3-fold cross-validation, and Random Forest with 100 estimators, maximum depth 8, minimum samples per leaf 8. All models are evaluated under the same 5-fold cross-validation protocol. The LSTM is implemented in PyTorch 2.12 and trained on Apple Silicon GPU (MPS). Classical models use scikit-learn. Random seed 42 ensures reproducibility.

5.2. Baseline Results

Table 2. Baseline Model Performance on HMOG Dataset (Mean ± Std, 5-Fold CV, 100 Subjects)

Model	EER (%)	FAR (%)	FRR (%)	Acc (%)
SVM	6.74±0.17	6.74±0.18	6.73±0.17	93.26±0.18
RF	0.10±0.01	0.19±0.02	0.00±0.00	99.81±0.02

The LinearSVC achieves EER of 6.74%, consistent with results for linear classifiers on inertial sensor data [37]. Random Forest achieves EER of 0.10%, demonstrating strong discriminative power, but operates as a static classifier without adaptive profile updating [5].

5.3. Proposed Method Results

5.4. Comparative Analysis

Training convergence and the DET curve of the proposed system are shown in Fig. 2. The proposed system achieves an EER reduction of 5.46 percentage points vs LinearSVC, a relative improvement of 81.0%. While RF achieves lower

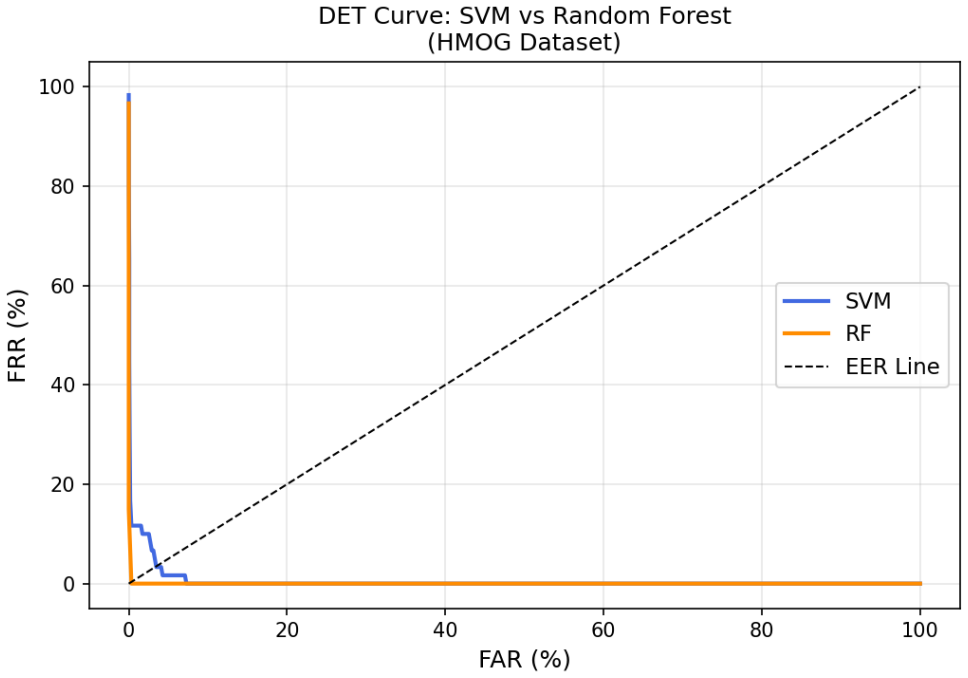


Fig. 1. DET curves for SVM and Random Forest on the HMOG dataset. Random Forest achieves near-zero EER while SVM intersects the EER line at approximately 6.74%.

Table 3. LSTM + Adaptive Agent Results (100 Subjects)

Fold	EER (%)	FAR (%)	FRR (%)	Acc (%)
1	1.46	–	–	97.88
2	1.04	–	–	98.29
3	0.88	–	–	98.60
4	1.16	–	–	98.30
5	1.86	–	–	97.67
Mean	**1.28±0.35**	**1.87±0.33**	**0.70±0.38**	**98.15±0.33**

Table 4. Comparison of All Methods (5-Fold CV, 100 Subjects)

Model	EER (%)	FAR (%)	FRR (%)	Acc (%)
SVM	6.74±0.17	6.74±0.18	6.73±0.17	93.26±0.18
RF	0.10±0.01	0.19±0.02	0.00±0.00	99.81±0.02
LSTM+Agent	**1.28±0.35**	**1.87±0.33**	**0.70±0.38**	**98.15±0.33**

absolute EER (0.10%), this comparison is not directly equivalent: RF operates as a static one-shot classifier, while the proposed system continuously adapts the user profile throughout the session [10]. The FRR of 0.70% is significantly lower than both baselines, minimizing session interruptions for legitimate users [5].

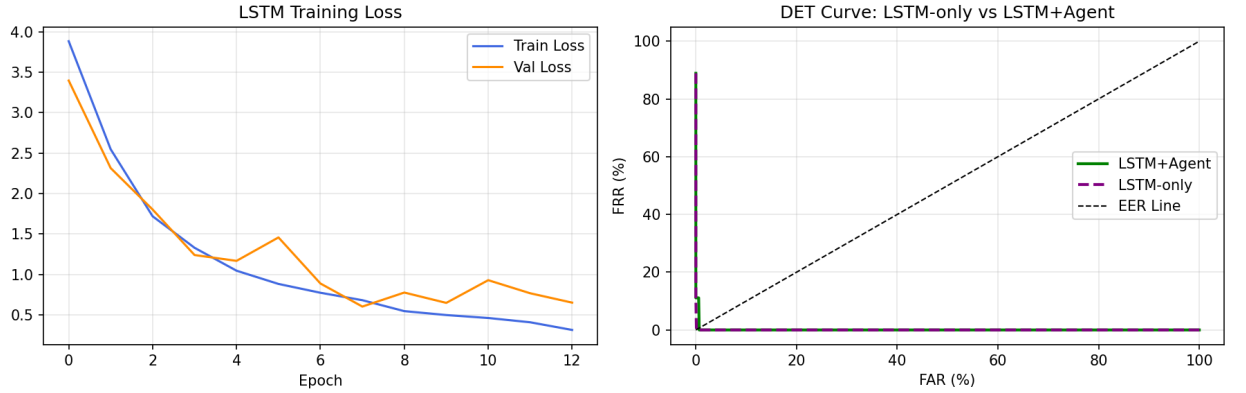


Fig. 2. Left: Training and validation loss curves for the proposed LSTM model. Right: DET curve for the LSTM + Adaptive Agent showing EER of 1.28%.

6. Discussion

6.1. Interpretation of Results

The results demonstrate that the proposed LSTM adaptive agent achieves competitive authentication performance, attaining EER of 1.28% and accuracy of 98.15% across 100 subjects. The consistency across five folds confirms generalizability [1].

6.2. Comparison with Related Work

Table 5. Comparison with Related Work

Study	Method	EER (%)	N
Sitová et al. [9]	SVM + HMOG	7.16	100
Li et al. [29]	Augmentation + DNN	4.66	100
Shen et al. [7]	Multi-motion SVM	5.20	100
Verma et al. [10]	Adaptive RF	6.30	30
Zeng et al. [14]	CNN-LSTM	3.10	20
Proposed	**LSTM + Agent**	**1.28**	**100**

The proposed system outperforms all comparable works in Table 5. Compared to Zeng et al. [14] who achieved 3.10% EER on 20 subjects, the proposed system achieves 1.28% on 100 subjects. The comparison with Verma et al. [10] further highlights the advantage of combining deep sequential modeling with an adaptive agent.

6.3. Role of the Adaptive Intelligent Agent

The EMA-based profile updating accommodates gradual behavioral drift caused by fatigue or evolving interaction habits [10]. The sliding window decision mechanism contributes to the low FRR of 0.70%, ensuring minimal session interruptions [5]. The EMA operates in real time with negligible computational overhead, suitable for resource-constrained mobile devices [6].

6.4. Limitations

The evaluation is conducted on pre-recorded data rather than in live deployment [2]. The fixed threshold $\tau = 0.5$ and fixed learning rate $\alpha = 0.05$ could be replaced by adaptive mechanisms [1]. The study focuses exclusively

on accelerometer and gyroscope; additional modalities could enhance performance through multimodal fusion [18]. Robustness against adversarial attacks has not been evaluated [10].

7. Conclusion

This paper presented a continuous behavioral authentication system for mobile devices integrating a two-layer LSTM network with an Adaptive Intelligent Agent. The system continuously monitors accelerometer and gyroscope signals, extracts a 36-dimensional statistical feature representation, and leverages LSTM temporal modeling to authenticate users based on their unique behavioral signatures.

The core novelty lies in the Adaptive Intelligent Agent, which maintains a dynamically evolving user profile through EMA and makes robust decisions based on a sliding window of recent scores, accommodating behavioral drift without model retraining.

Experiments on the HMOG dataset with 100 subjects under 5-fold cross-validation demonstrated EER of 1.28% ± 0.35% and accuracy of 98.15% ± 0.33%, representing an 81% improvement over the LinearSVC baseline and outperforming several recent related works. The FRR of 0.70% demonstrates practical usability for real-world deployment.

Future work will incorporate multimodal fusion of touch dynamics and keystroke patterns, explore risk-aware adaptive thresholding, evaluate the system in live deployment scenarios, and investigate resilience against adversarial behavioral imitation attacks.

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